

Traffic Management Effects of Variable Speed Limit System on a German Autobahn

Empirical Assessment Before and After System Implementation

Simone Weikl, Klaus Bogenberger, and Robert L. Bertini

An empirical study of the traffic flow effects of a variable speed limit (VSL) system on a three-lane German autobahn is presented. The study integrated loop detector data and VSL and driver information provided by overhead dynamic message signs along a 16.3-km section of Autobahn A99 near Munich, Germany. Unlike most VSL deployments, the data sources allowed for a with-and-without analysis to reflect traffic flow and driver behavior in the presence and in the absence of the VSL system. In addition to speed, the primary features of interest were the spatial-temporal extent of the queue (congestion), flow changes caused by identified bottlenecks, the distribution of flow across lanes, the percentage of trucks per lane, and the flow homogeneity between lanes. The analysis recorded bottleneck flow reductions that were balanced across all lanes with an active VSL system. With an inactive VSL system, the flow reductions were slightly lower and occurred primarily in the center and shoulder lanes; there was a small flow increase in the median lane. Further examination indicated that the gain in homogeneity and safety with the system in place was at the cost of capacity. Possible reasons that might explain the differences between the scenarios with and without the VSL system had their basis in the ban against passing by trucks, which was in effect when the driver information system was active.

Active traffic management systems are being implemented around the world with the use of measured data, advanced technology, and proactive management strategies to maximize safety, flow, reliability, and sustainability within freeway and arterial corridors. In some cases, formal evaluations are conducted before, during, or after implementation, but it is relatively rare to find high-resolution measured data available when a system is installed but not yet in operation. This paper presents a unique with-and-without view of a variable speed limit (VSL) and driver information system that has been installed in both directions on a 16.3 km, primarily three-lane section of Autobahn A99 in Munich, Germany (Figure 1). This system uses overhead dynamic message signs spaced every

1 to 2 km, designed to postpone or prevent freeway breakdown, dampen upstream-moving shock waves, harmonize flow and speed across lanes during peak periods, and reduce vehicle crashes. The system combines the use of real-time traffic information with data provided via inductive loop detectors, cameras for speed enforcement, and remote weather information systems to evaluate traffic, weather, and roadway conditions to determine a recommended speed or advisory for a particular autobahn segment.

Earlier studies mostly focused on the safety impacts of VSL systems (3, 4). It has been shown that German VSL systems have contributed to decreased crash rates by as much as 20% (3). Other research has analyzed the dynamics of VSL systems that surrounded bottlenecks on a German autobahn (5). Speed limit and information messages have been compared with actual traffic dynamics, and the analysis has found a strong correlation. Other studies examined the dynamics of traffic flow and physical freeway infrastructure before the occurrence of bottlenecks on freeways with a VSL system (6, 7). These studies measured data only after the VSL system was in operation.

The question whether these traffic management investments increase the capacity of the freeway has not yet been answered definitively. Some studies have indicated that VSL systems can increase freeway capacity by up to 10% (8). Studies of other European freeways showed that a VSL system did not substantially improve freeway performance (9, 10). This paper describes an empirical with-and-without analysis to measure the impacts of the VSL system on traffic flow characteristics in a corridor. The analysis used archived loop detector data from a period when the VSL system was not activated, and loop data and actual VSL and driver information messages when the VSL system was in operation. With high-resolution, lane-by-lane data for two comparable periods, it was possible to carefully and systematically analyze differences between traffic flow features with and without the VSL system. Freeway bottleneck features that were analyzed included speed and flow on a per lane and per vehicle basis, prequeue flow and bottleneck outflow, percentage of trucks per lane, shock velocities, and homogeneities between lanes.

VSL SYSTEM, SITE, AND DATA

The A99 VSL and driver information system used algorithms to determine the most appropriate speed limit based on three control strategies. These include incident detection, harmonization, and weather detection (8). Lane harmonization aims to control dense but still flowing traffic through the balance of speed and flow across

S. Weikl and K. Bogenberger, Department of Civil Engineering and Environmental Sciences, Munich University of the Federal Armed Forces, Werner-Heisenberg-Weg 39, 85577 Neubiberg, Germany. R. L. Bertini, Department of Civil and Environmental Engineering, Portland State University, P.O. Box 751, Portland, OR 97207-0751. Corresponding author: S. Weikl, simone.weikl@unibw.de.

Transportation Research Record: Journal of the Transportation Research Board, No. 2380, Transportation Research Board of the National Academies, Washington, D.C., 2013, pp. 48–60.
DOI: 10.3141/2380-06

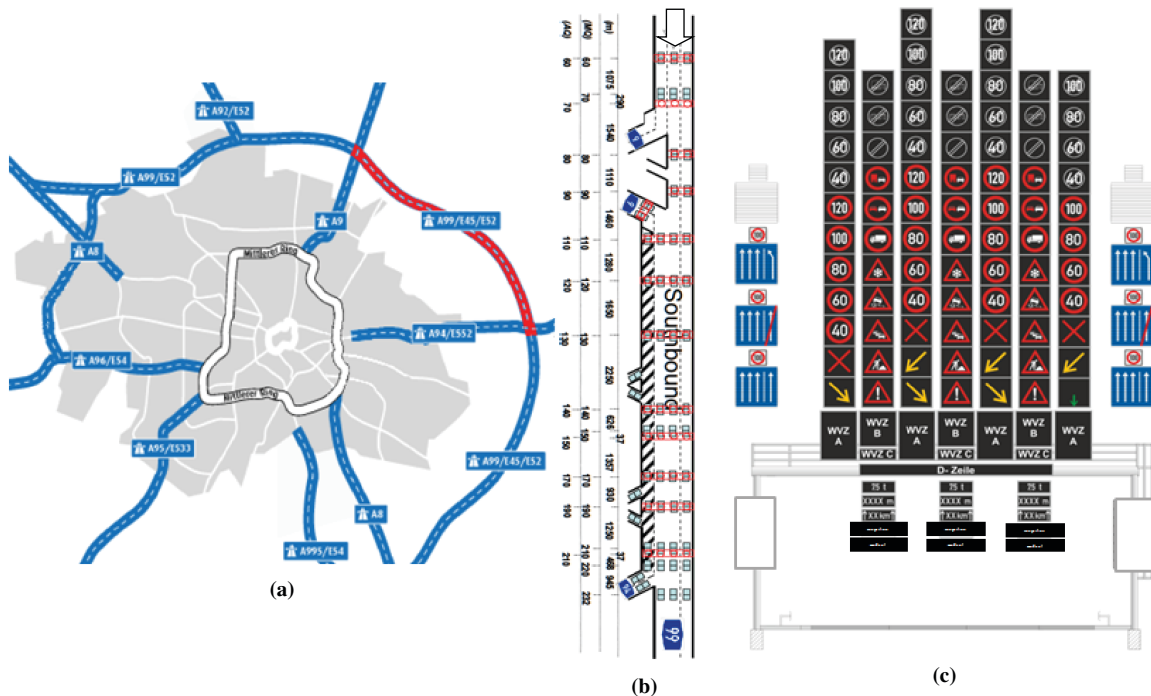


FIGURE 1 Views of VSL system on 16.3-km, three-lane section of Autobahn A99, Munich: (a) study area (1), (b) road network of study area (southbound A99), and (c) possible message signs of VSL system (2).

all lanes and to postpone or prevent a breakdown. The need for lane homogeneity is heightened in Germany, where driving rules include separate speed limits for trucks, and require trucks and slow-moving vehicles to remain in the right lane, with the left lane reserved for passing. The VSL system compared measured values to thresholds and deployed a recommended speed, a warning, or both. The values were calculated for each section on the basis of data measured across all lanes. Those data included mean speed, density, free speed, and maximum density. In most cases, lanes were not considered separately by the algorithms (11). When a critical state was identified, the system reduced the speed limit upstream and warned approaching drivers of the potential for a breakdown or the existence of one downstream already. Automated speed enforcement with cameras was incorporated into the overhead gantries to ensure high-speed compliance rates. The cameras were deployed on a random basis only, however. Drivers did not know when and where the speed limit would be enforced. The impact of enforcement on the VSL effect was not examined in this study. The ban on overtaking by trucks was not enforced for cost reasons.

The study reported in this paper used a combination of archived loop detector and VSL system data to measure traffic flow features on the segment shown in Figure 2. Fourteen dual-loop detector stations were located in the travel lanes, and six dual-loop detectors were located on the on- and off-ramps. Each detector provided separate count, occupancy, and time–mean speed data at 1-min intervals for autos and trucks. When the VSL system was in place, VSL and messages displayed on the 13 variable-message signs also were available. Unique to this study, when the VSL system was not operative, detector information was available. Archived inductive loop detector data were available for both directions of the freeway for 46 days (October 10–25 and November 1–30, 2003) while VSL data were available for 35 days (October 10–24 and

November 11–30, 2003). The VSL system was not operative on October 25, or November 1–10, 2003. Data for weekends were excluded from this analysis. Data were considered for 25 weekdays with the VSL system in operation, and for 6 weekdays without. Weather was not a substantial variable. On most days, conditions were foggy with light rain.

A statistical analysis of the time series according to similar demands, and other factors, was not part of this study. Because the total sample was limited for the days with and without the VSL system, the only possible criteria for the analysis were weekdays and no atypical incidents.

BOTTLENECK DIAGNOSIS

The analysis focused on differences in measured traffic flow features between the prequeue (uncongested) period and the queue discharge period. To map the spatial–temporal extent of the queues upstream of bottlenecks, their activation and deactivation times were identified. Knowledge of the bottleneck locations was essential to measure flow features. Five congested periods on 4 days were examined in detail, 18 bottlenecks with queues were identified in time and space, and the propagations of flow features were traced. The study reported on several features that were reproducible across all study days.

For example, the bottleneck identification method was shown for October 16, 2003, when the VSL system was operative, and displayed warnings and speed limits during the congested period (Figure 2b and c). The bottleneck location was estimated through a view of the speed plot (Figure 2a), which showed speeds averaged across all lanes for each 1-min interval, with time as the x-axis and distance as the y-axis, and with speed (combined autos and trucks) plotted in color. Green indicated high speeds, and yellow and red

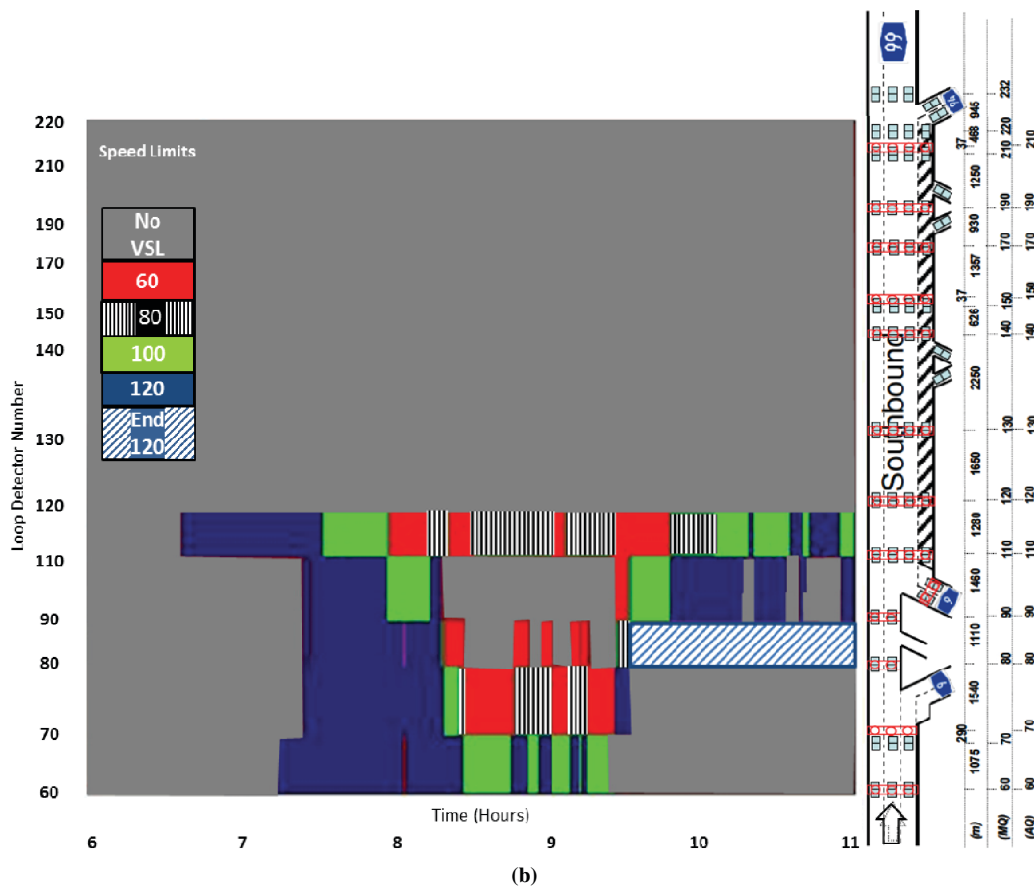
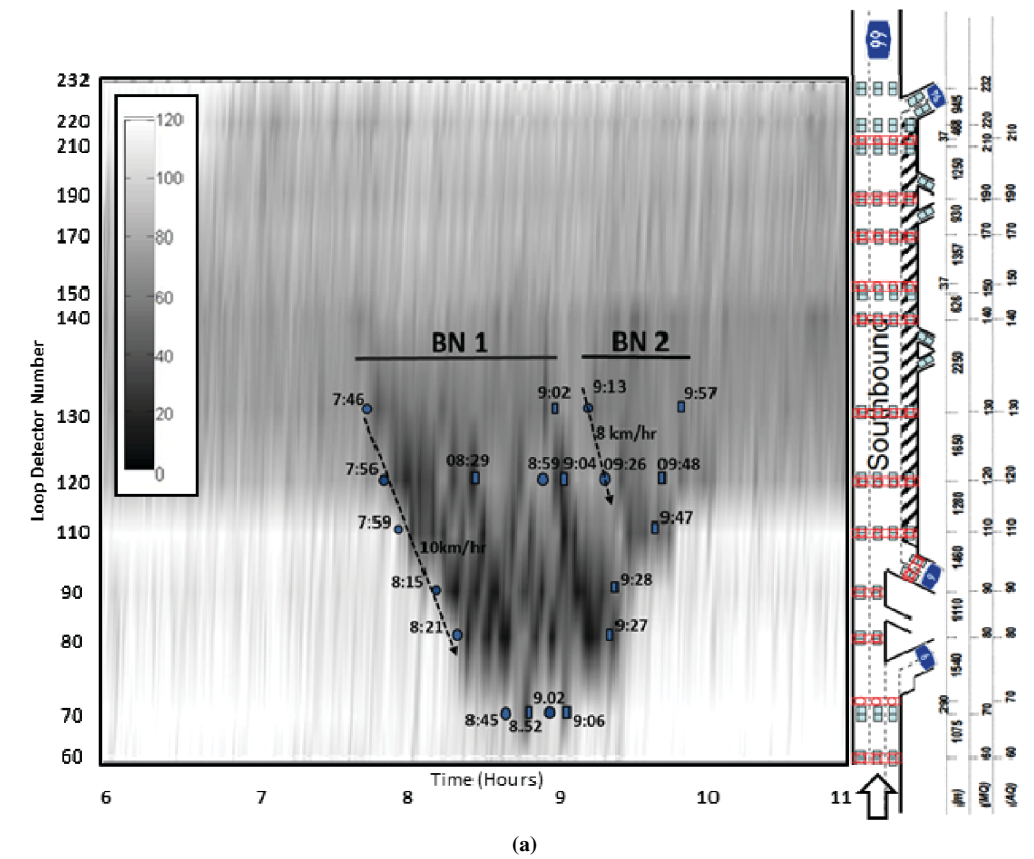


FIGURE 2 Traffic flow features measured on southbound segment of A99: (a) plot of speed data for October 16, 2003, and (b) speed limits on October 16, 2003.

(continued)

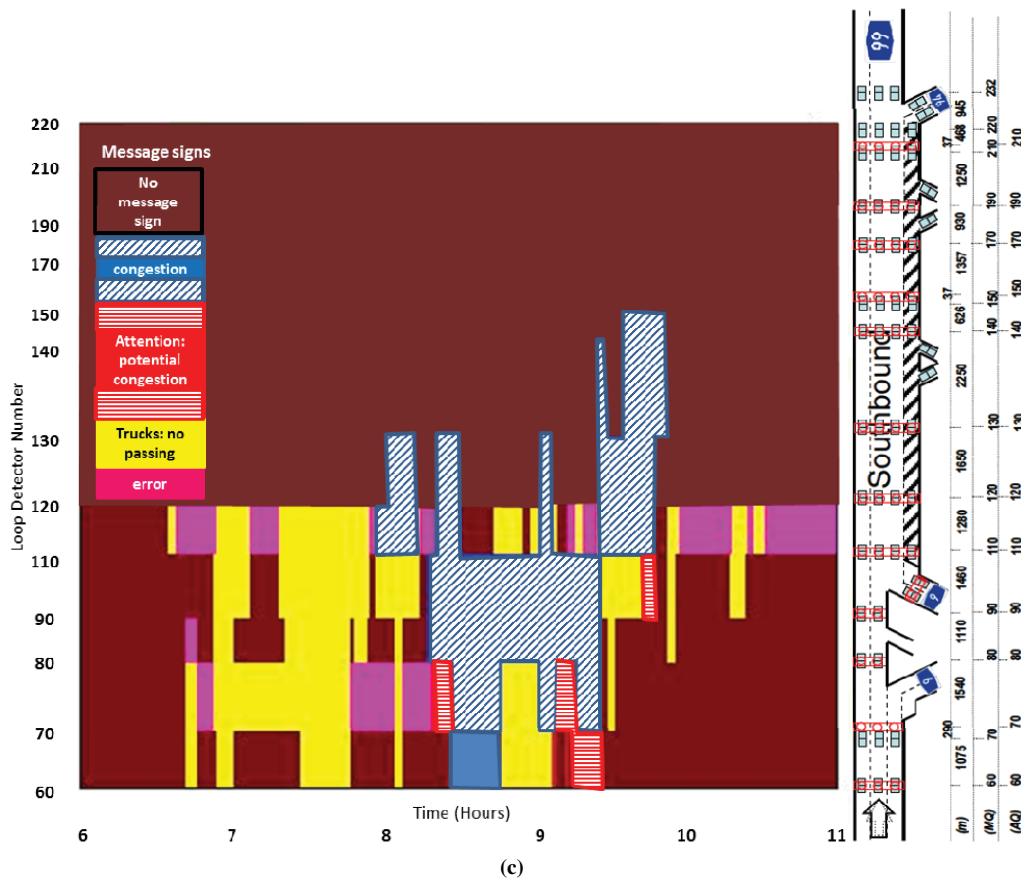


FIGURE 2 (continued) Traffic flow features measured on southbound segment of A99: (c) message signs on October 16, 2003.

represented lower speeds. Several speed disturbances occurred between 7:45 and 10:00. Two bottlenecks diagnosed on October 16, 2003, labeled BN1 and BN2, were mapped in time and space (Figure 2a). From the plot, it appeared that bottleneck activations BN1 and BN2 occurred between Detectors 130 and 140. Queues propagated upstream of Detector 130, and queue discharge characteristics could thus be measured at downstream Detector 140.

Identification of bottleneck locations, activation and deactivation times, and diagnosis of their features are well established (12–16). These techniques were used in this study to carefully distinguish between queued and unqueued states, and to ensure that flow measurements were made at the correct locations and times. Earlier studies used these techniques with data from I-405 in Orange County, California (17), and from Autobahn 9 in Munich (5, 18). They provided an easily followed template for this analysis. Figure 3a shows transformed, oblique curves of cumulative vehicle count $[N(x, t)]$ for Detectors 130 to 170 from 7:37 until 8:24. With the 1-min count data measured across all lanes, piecewise linear approximations of the cumulative counts were constructed so that the slope of the unaltered $N(x, t)$ would be the flow past location x at any time t . The counts for each curve began ($N = 0$) so that each curve described the same collection of vehicles. The curve at 130 included on-ramp counts but not off-ramp counts so that vehicle conservation was maintained. Horizontal and vertical separations between $N(x, t)$ would have been the travel times and vehicle accumulations between measurement locations, respectively (16). In Figure 3a, the curves

were altered through the shift of each upstream curve to the right by the free-flow travel time from its location to Detector 170. Vertical displacements between the curves showed the excess vehicle accumulation between detector pairs, and the horizontal distance between curves were the excess travel times (delay) between the measurement locations (16). To amplify the curves' features further, an oblique scaling rate (q_0) was applied to the $N(x, t)$, which revealed the times when notable flow changes occurred. Figure 3a has an amplified, vertical scale, so that changes in flow can be observed clearly (in this and subsequent figures a relative vertical scale is shown on the left axis). The curves in Figure 3a remained superimposed until 7:47, which indicated that traffic flowed freely between stations. At 7:47, the figure showed excess accumulation between Stations 130 and 140, as indicated by the divergence of the oblique $N(x, t)$. A similar divergence of Curves 150 and 170 (and flow reductions) marked the passage of the backward-moving queue. The oblique $N(x, t)$ for Stations 140, 150, and 170 remained approximately superimposed. This activation was labeled as BN1, which lasted until 9:02.

Figure 3b traces this queue further upstream with transformed curves from Stations 130, 120, and 110, for which vehicle conservation was possible between the three locations. The queue passed Detector 110 at 7:59, Detector 90 at 8:15, and Detector 80 at 8:21. The curves of upstream Detectors 60 and 70 remained superimposed. This finding indicated free-flowing conditions between those two detectors. Therefore, the spatial end of the queue was located between Detectors 70 and 80.

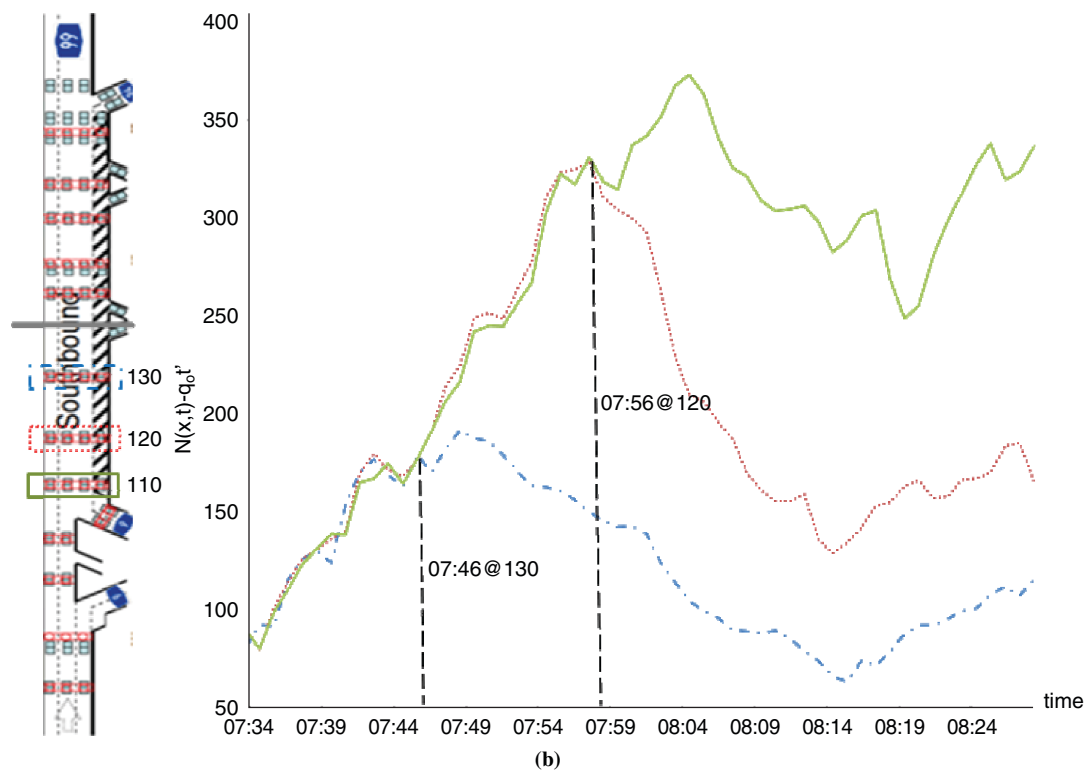
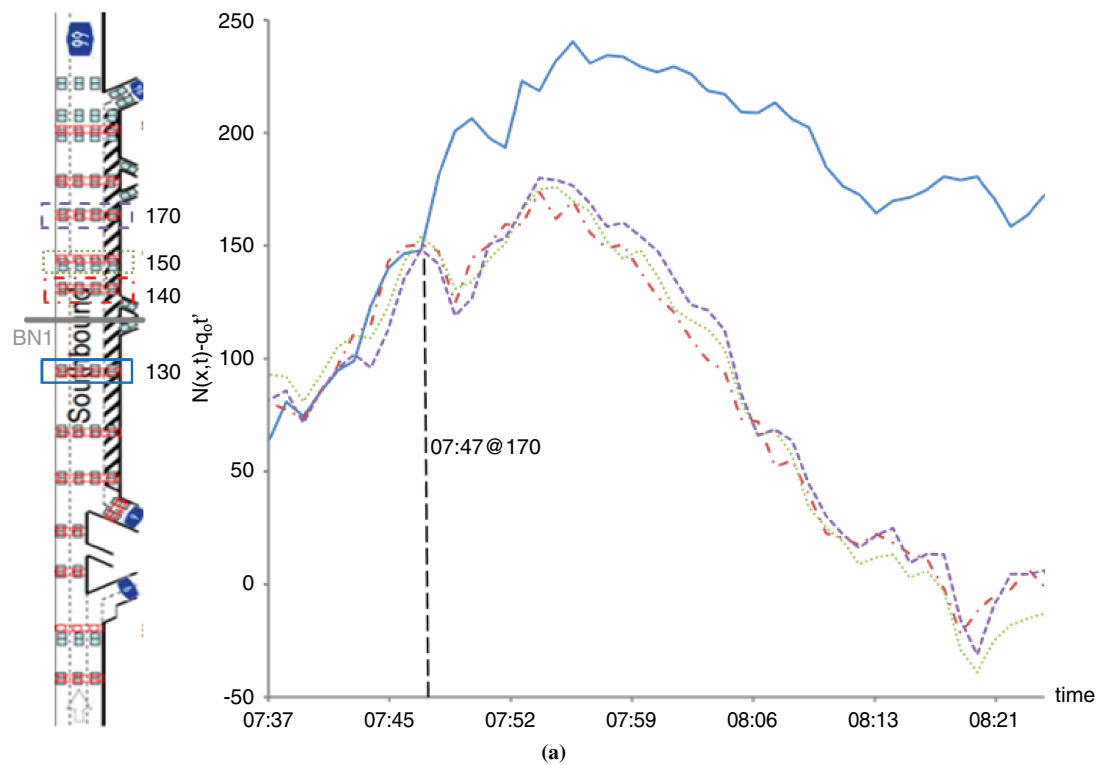


FIGURE 3 Transformed cumulative curves of vehicle count: (a) Detectors 130–170 and (b) Detectors 110–130.

A second bottleneck (BN2) was analyzed, and its activation and deactivation times and spatial extent are shown in Figure 3a. The activation and deactivation times were confirmed with transformed, average speed curves (not shown here). A sharp decrease in slope signaled a speed decrease and confirmed the queue's passage. When the bottleneck was deactivated, the recovery was mapped by measured increases in average speed.

TRAFFIC FLOW FEATURE MEASUREMENT

After the spatial-temporal extent of the queues for each detector was identified, an analysis was done of the bottleneck prequeue flow, the queue discharge flow, and the resulting change in flow and capacity (Figure 4). Measurements were made on a per lane and per vehicle basis at the first detector downstream of the bottleneck, which exhibited free-flowing conditions both before and during the bottleneck activations. Prequeue flow was the prevailing average flow in the last period, with a nearly constant slope before bottleneck activation (e.g., 4,790 vph for BN1 and 4,500 vph for BN2). The queue discharge flow was the average flow during the period when the bottleneck was active (e.g., 4,300 vph for BN1 and 4,350 vph for BN2).

The shock velocity was calculated from the spatial and temporal extent of the queue. The accuracy of shock velocities was limited by the 1-min resolution of the data (16). The percentage of trucks in the overall flow was analyzed on a by-lane basis. Earlier research showed that a truck ban and a VSL system had a profound impact on the lane flow distributions on a French motorway (19). In the study reported here, the flow homogeneity across lanes was investigated through a plot of the distribution of vehicle flow between lanes. The focus was on the correlation of the lane flow distribution and the total flow of the freeway.

RESULTS

Congestion Forms

In addition to the measurement of traffic flow features, speed contour plots for all days and directions were analyzed, and the queues were assigned into one of three categories: (a) stop-and-go traffic, (b) single shock waves, and (c) wide jams (20). This categorization (Table 1) had its basis in an analysis of more than 600 congestion forms and was used to identify differences with respect to the queues in this analysis. On more than half of the days that the VSL system was operative, the congestion form was a wide jam. Stop-and-go waves were also dominant, with a percentage share of 36%, whereas single shock waves were exceptions. When the VSL system was not operative, stop-and-go traffic occurred more frequently than large queues. The data set for the days without an activated VSL system was small compared with the set when the system was operative. Future studies are needed to confirm these findings.

Prequeue Flow, Queue Discharge Flow, and Flow Change

Table 2 summarizes the traffic flow features for the 18 bottlenecks analyzed, including VSL status, date, location, activation, and deactivation time, prequeue and queue discharge flows (overall and by lane), flow change, and shock velocity. Table 3 shows the mean values for VSL on and VSL off. An investigation of those average values and features produced the following results.

With an active VSL system, the flow reduction of approximately 4% was evenly distributed across all three lanes (4%, 3%, and 4% respectively). When the VSL system was not operative, the flow

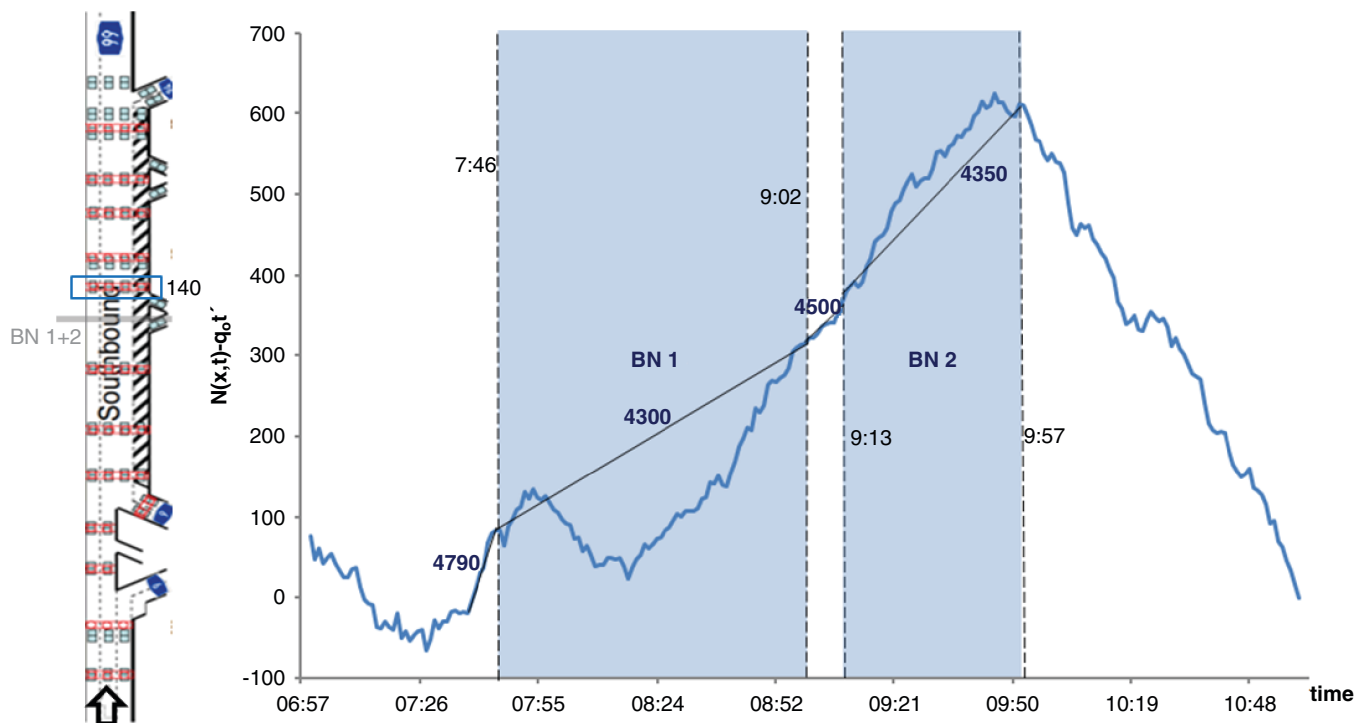
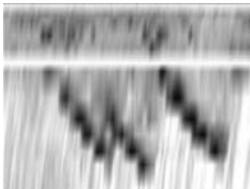


FIGURE 4 Measurements of prequeue flow and queue-discharge flow at first detector downstream of bottleneck.

TABLE 1 Occurrence of Congestion Forms

Queue Categorization	Example from Test Site	Characteristics	VSL On	VSL Off
Single shock wave		Narrow red stripe Vehicle has to brake heavily once Shock wave seems to appear from nowhere High decrease of speed from 120 to 20 km/h Spatial extent: ca. 1 km Loss of time: ca. 3 min	10%	6%
Stop and go		Several narrow red shock waves (separated by green areas with free flow) Frequent braking—drive away—braking Speed varies strongly between 0 and 80 km/h Spatial extent: ca. 12 km Loss of time: in overall ca. 21 min	36%	50%
Wide jam		Large with congestion colors (red, orange, yellow) Definite bottleneck Uniform speed between 0 and 40 km/h Spatial extent: ca. 3 km Loss of time by low average speed: ca. 15 min	54%	44%
Mega jam	No observation	Very large area with congestion colors (red, orange, yellow) Spatial extent: ≥ 30 km Very high loss of time Speed varies strongly between 0 and 80 km/h Spatial extent: ca. 30 km Loss of time: in overall ca. 38 min	0%	0%

NOTE: Values in boldface indicate dominant queue types.

reduction of 3% was slightly lower than when the system was active. A small flow gain of 1% was detected in the median lane when pre-queue and queue discharge flows were compared. For the center and median lanes, the reduction in flow was almost evenly distributed but was 5% and 6% higher than when the system was active. In addition, when the system was off, total flow values were higher during the prequeue period and during the queue discharge. The flow difference between system on and system off was 150 vph.

Lane Flow Distribution

The flow distribution by lane showed an underuse of the right lane (25% share) in both the system on and system off cases. Table 2 does not include the vehicle equivalents for trucks, but only the raw number of all the vehicles that passed the detector. Thus the high percentage of trucks in the right lane led to the low percentage share of the shoulder lane. Nevertheless, differences occurred between the two cases with respect to the use of the center and the median lanes. When the system was on, the center and median lanes were used equally, with averages of 38% and 37%. An imbalance occurred when the system was off, with use rates of 40% and 34%, respectively.

For the flow homogeneity analysis, the distribution of vehicle flow between lanes was not normalized through a conversion of the volume to passenger car equivalents. The focus was on the correlation of the

lane flow distribution and the total flow of the freeway. For instance, the effects of lane changing and the differences between the lanes were more visible without the conversion. However, to ensure completeness, the percentage of trucks per lane was analyzed separately. The lane flow distribution was calculated for passenger car equivalents (Table 3). With this normalization, the flow in the right lane was higher, but the right lane was still underused.

Spatial-Temporal Extent and Queue Propagation

The bottleneck locations were identified with the methodology described earlier. With use of the VSL system, warnings, or both, a bottleneck was predominantly located between Detectors 130 and 140. This location was in the direct vicinity of an exit and an entrance ramp (Figure 1). By contrast, the bottleneck, on average, shifted downstream between Detectors 140 and 170 when the VSL system was inactive. Thus the bottlenecks occurred farther downstream of the exit and entrance ramps when traffic flow was not influenced externally.

A spatial-temporal analysis of the queued regions showed further deviations between the two cases. When the VSL system was on, the congested region consisted of one wide jam, whereas when the system was off, numerous jams occurred, which caused stop-and-go traffic. It was assumed that the safety level was higher when the system was on, because wide jams pose a lower risk for vehicle

TABLE 2 Bottleneck Summary: Prequeue Flow, Queue Discharge Flow, and Flow Change

VSL Status	Date	BN Location	Begin	End	Prequeue Flow (vph)				Queue Discharge Flow (vph)				Flow Change (%)				Shock Velocity (km/h)
					Overall	Per Lane			Overall	Per Lane			Overall	Per Lane			
						Median	Center	Right		Median	Center	Right		Median	Center	Right	
On	10/16/03	130–140	07:46	09:02	4,790	1,810	1,830	1,150	4,300	1,620	1,610	1,080	–10	–10	–12	–6	10
On	10/16/03	130–140	09:13	09:57	4,500	1,680	1,670	1,150	4,350	1,720	1,530	1,100	–3	2	–8	–4	8
On	10/20/03	140–150	07:30	07:36	4,680	1,620	1,740	1,320	4,650	1,580	1,830	1,240	–1	–2	5	–6	na
On	10/20/03	130–140	07:56	08:42	4,610	1,850	1,700	1,060	4,600	1,810	1,690	1,110	0	–2	–1	5	12
On	10/20/03	130–140	08:52	09:07	4,880	1,910	1,760	1,210	4,730	1,880	1,740	1,110	–3	–2	–1	–8	17
On	10/20/03	130–140	09:26	09:50	4,820	1,840	1,730	1,240	4,690	1,810	1,720	1,160	–3	–2	–1	–6	na
On	10/20/03	210–220	10:07	10:27	4,360	1,410	1,610	1,340	4,030	1,380	1,590	1,060	–8	–2	–1	–21	na
On	10/20/03	90–110	07:43	09:45	4,990	2,150	1,760	1,080	4,660	1,810	1,630	1,210	–7	–16	–7	12	8
Off	11/03/03	190–210	07:48	08:13	4,400	1,860	1,590	950	4,380	1,920	1,590	870	0	3	0	–8	na
Off	11/03/03	130–140	07:56	08:04	5,180	2,060	1,890	1,230	4,920	1,890	1,990	1,030	–5	–8	5	–16	15
Off	11/03/03	120–130	08:18	08:24	4,910	1,930	1,800	1,180	5,100	2,050	1,870	1,180	4	6	4	0	21
Off	11/03/03	210–220	08:30	08:35	4,700	2,090	1,510	1,100	4,690	2,180	1,480	1,030	0	4	–2	–6	12
Off	11/03/03	210–220	17:40	18:14	4,960	1,940	1,690	1,330	4,880	2,000	1,520	1,360	–2	3	–10	2	na
Off	11/03/03	190–210	17:38	17:55	5,140	2,140	1,860	1,140	4,570	2,090	1,500	980	–11	–2	–19	–14	10
Off	11/05/03	140–170	07:58	08:07	4,860	1,880	1,520	1,450	4,630	1,790	1,400	1,440	–5	–5	–8	–1	16
Off	11/05/03	140–170	08:31	08:49	4,850	1,750	1,600	1,500	4,750	1,810	1,530	1,420	–2	3	–4	–5	12
Off	11/05/03	140–170	09:01	09:08	4,680	1,620	1,470	1,590	4,690	1,790	1,470	1,430	0	10	0	–10	33
Off	11/05/03	140–170	09:23	09:32	4,570	1,790	1,490	1,290	4,310	1,730	1,270	1,310	–6	–3	–15	2	na

NOTE: na = not applicable.

TABLE 3 Mean Value of Bottleneck Summary with VSL On and VSL Off

		Prequeue Flow (vph)				Queue Discharge Flow (vph)				Flow Change (%)				Shock Velocity (km/h)
VSL Status	BN Location	Overall	Per Lane			Overall	Per Lane			Overall	Per Lane			
			Median	Center	Right		Median	Center	Right		Median	Center	Right	
On	130–140	4,700	1,780	1,730	1,190	4,500	1,700	1,670	1,130	−4	−4	−3	−4	11
	Lane flow distribution		38%	37%	25%		38%	37%	25%	na	na	na	na	na
	Passenger car equivalents		34%	34%	31%		34%	34%	31%	na	na	na	na	na
Off	140–170	4,830	1,910	1,640	1,280	4,690	1,920	1,560	1,210	−3	1	−5	−6	17
	Lane flow distribution		40%	34%	27%		41%	33%	26%	na	na	na	na	na
	Passenger car equivalents		35%	34%	30%		37%	33%	30%	na	na	na	na	na

crashes than do frequent encounters of stop-and-go traffic. The spatial-temporal diffusion of the queues was similar in both cases. On October 16, 2003, for example, the queue that emerged in the presence of an active VSL system was 9.5 km in length and lasted 1.5 h. The aggregation of the numerous jams that emerged on November 5, 2003, without the VSL system in active mode, was slightly longer in space and time (11 km and 2 h). Studies are needed to confirm these findings through a search for average values for the spatial-temporal extent.

If the VSL system was not operating, the queue appeared to propagate with substantially higher speed. It appeared that the desired damping effect of the VSL system was measurable within the data. The mean shock speed difference between system on and system off was 6 km/h. The higher shock velocity presumably reduced safety and created a higher risk for vehicle crashes at the spatial end of the queue.

Percentage of Trucks per Lane

Another important feature was the percentage of trucks per lane in the overall flow. This was calculated for the 18 bottlenecks (Table 2). In both cases, almost no difference occurred between the prequeue and the queue discharge percentage of trucks per lane. There were, however, notable differences in the percentage of trucks when the system was on and when it was off. When the VSL system was operative, more than half of the vehicles in the shoulder lane were trucks (56%). Some trucks were in the center lane (5%), but there were almost none in the median lane (1%). When the VSL system was not operative, more trucks were in the center lane. Almost a quarter of all of the vehicles in the center lane were trucks (21%). Slightly more trucks used the median lane (2%). As a result of these values, there were fewer trucks in the shoulder lane (43%). Clearly, these findings demonstrated that truck operators were more compliant when the VSL system was operative.

Homogeneity Between Lanes

The homogeneity between lanes was analyzed also. The ratio of the flow per lane to the total freeway flow was calculated and plotted at three locations. First, a detector upstream of the congested area (free-flowing conditions) was considered. Thus the behavior of drivers just before they reached the spatial end of the queue could be reconstructed. Second, a detector in the congested area was used. Finally, a detector downstream of the congested area was studied for information about the behavior of the drivers when they operated outside of the queue. Results were compared for the system on and system off cases. The analysis of different days on and off showed similar results. Figure 5 shows sample diagrams for 2 days, one with an operative VSL system and one without.

Without a VSL system, the lane flow distribution upstream of the congestion was relatively balanced, both for lower and higher total flow. The use of each lane was almost constant. The flow in the median lane was always higher than the flow in the center lane. The shoulder lane was underused. In the congested area, the situation was different. For lower total flow, more vehicles drove in the center lane than in the median lane. This distribution went into reverse with higher total flow, and thus there was a shift to the left. The shoulder lane was underused but constant. The situation at the downstream detector was similar to the one at the upstream detector, but the

underuse of the shoulder lane was slightly higher, and the difference between the shoulder lane and the two left lanes thus was higher. The lane flow distribution was nearly constant and did not change with the total flow.

With the VSL system, the situation was different. At the upstream detector, the median and center lane use was balanced. The difference in flow between these two lanes was smaller than when the VSL system was off. With higher total flow, the underuse of the shoulder lane was reduced, and there was even a small shift from left to right. These findings showed that the influence of the VSL system on driver behavior increased as the flow increased. The VSL system had a harmonizing effect, as desired. In the congested area, the flow in the shoulder lane was nearly constant, and the shoulder lane was always underused. The median and center lanes again were more homogeneous with higher total flow, and there was a shift from the median to the center lane. The homogenization thus was limited to the two left lanes in the congested area. With respect to the first detector downstream of the bottleneck, the situation was different. With lower total flow, the lane use was relatively balanced. The flow in the shoulder lane was even higher than the flow in the median lane. The flow in the center lane was almost constant and did not depend on the total flow. For higher total flow, there was a shift from the shoulder lane to the median lane. The shoulder lane was then as underused as at the detectors examined earlier. Nevertheless, the median lane was not as highly used as at the other detectors. In contrast to the situation when the system was off, the center lane was the dominant lane except for high total flows.

Possible Reasons for Empirical Findings

Given the differences in measured traffic flow features described earlier, it was concluded that it might be possible to identify reasons for the findings. The measured vehicular behavior was examined again after free-flowing conditions were reached. For this purpose, the average results of a data analysis of the following two categories of detectors were compared: the last detectors in the queued regime upstream of the bottleneck and the first detectors in the free-flowing conditions downstream of the bottleneck (Table 2). The analysis also included a comparison of the prequeue period and the queue discharge period.

VSL System On

Analysis of the two chosen categories of detectors when the system was on considered the final detectors in the queued area upstream of the bottleneck. The flow decreased 2% once the bottleneck occurred, with almost the same flow reductions in the median and the center lanes and the flow unchanged in the shoulder lane. The underuse of the shoulder lane decreased in the transition from the prequeue period to the queue period. Nevertheless, the shoulder lane was underused in both periods. The flows in the median and the center lanes were higher than the flow on the shoulder lane. In part, this result was due to the high percentage of trucks in the shoulder lane. The use of the median and the center lanes was relatively balanced. Moreover, the data analysis showed a high percentage of trucks in the shoulder lane. The center lane was used by some trucks, whereas there were almost no trucks in the median lane. The traffic situation for the congested period is shown in the left part of Figure 6a.

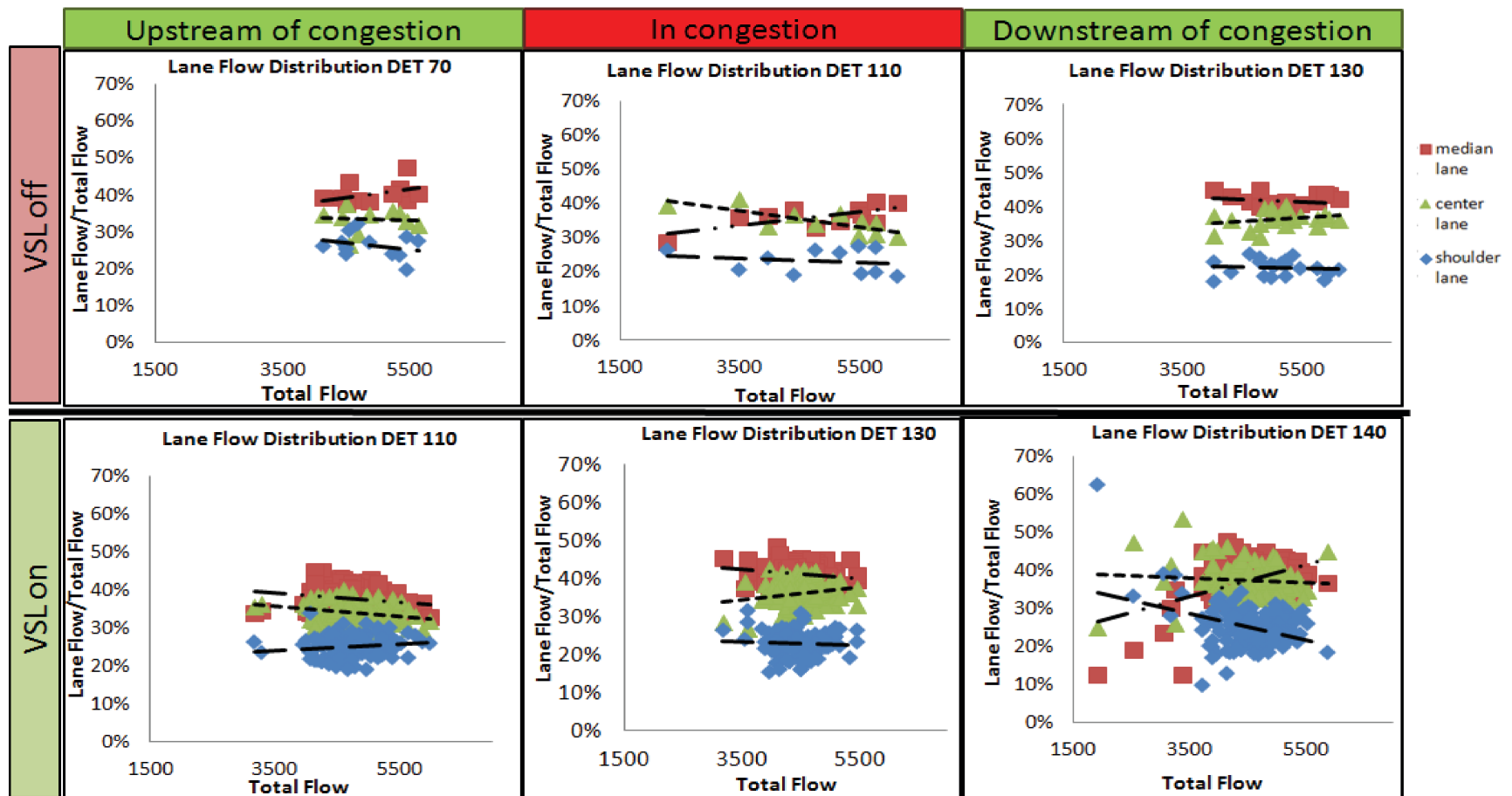


FIGURE 5 Lane flow distribution upstream of congestion, in congestion, and downstream of congestion (comparisons made when VSL system was on and off).

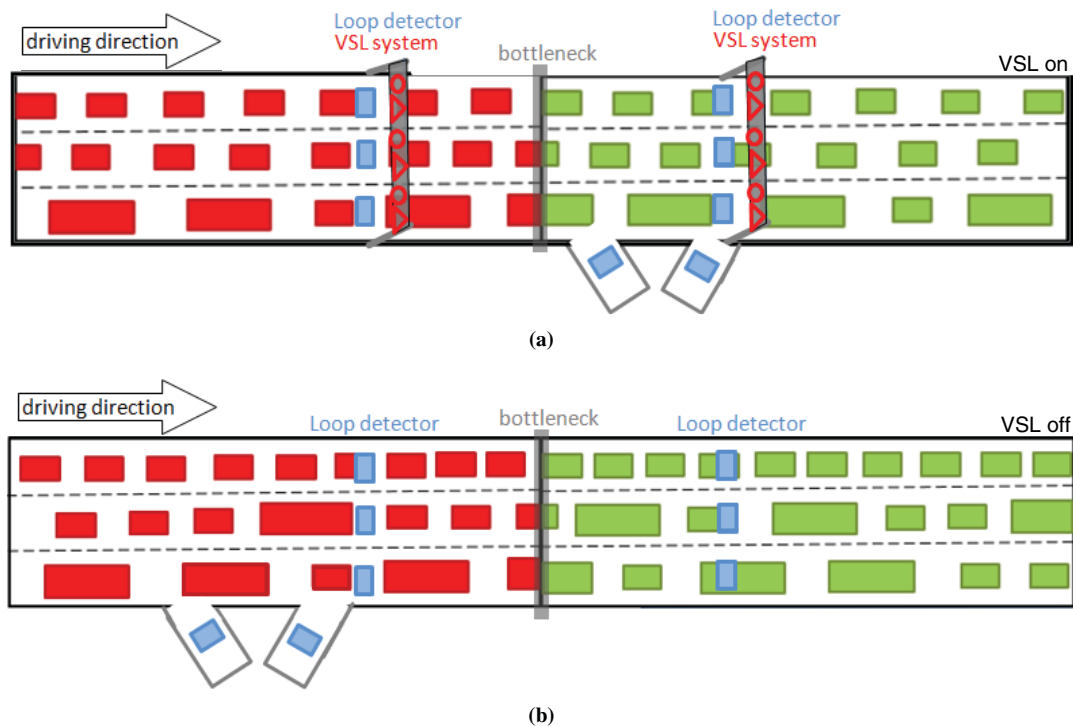


FIGURE 6 Possible behavior during drive out of queue: (a) with VSL system and (b) without VSL system.

The first detectors in the free-flowing conditions downstream of the bottleneck were examined—they revealed a flow reduction of 4%, with reductions in all lanes. The median and the shoulder lanes revealed higher flow reductions than the center lane. The lane flow distribution and the percentage of trucks in all three lanes were similar to those in the vicinity of the upstream detector. A slight increase in homogeneity and in the use of the shoulder lane occurred. This result might have been due to the higher number of trucks in the shoulder lane. The traffic situation for this category of detectors is shown in the right portion of Figure 6a.

Through a comparison of the results for the two detectors, driver behavior could be modeled between the detectors, and thus driver behavior out of the queue could be modeled as well. This behavior is illustrated in Figure 6a. In the examined case in which the VSL system was on, drivers were still influenced by the VSLs and warnings while they drove out of the queue. In most cases, VSL signs remained at the detector downstream of the bottleneck (Figure 2b). The drive out of the queue thus was relatively homogeneous, and even slightly more homogeneous than in the congested area, because more vehicles (especially trucks) drove in the right lane when traffic was freely flowing again, although there was an on-ramp with relatively high flow (Figure 7) in the vicinity of most of the detectors in this category. This finding might reflect decreased aggression and excitement on the part of the drivers once they reached the end of the congestion. Moreover, drivers might have assumed that the probability of a check on following distance, passing, or speed (and therefore a ticket) was higher when the VSL system was on, because detection measurements would be easy to obtain with the use of the VSL system infrastructure. Drivers thus drove more slowly, with a higher following distance, and made fewer passing maneuvers. Thus the drops in flow and capacity were relatively homogeneous across all lanes and higher than expected.

VSL System Off

The traffic flow features for the two chosen detectors when the VSL system was off are explained in detail. First, the analysis of the category of the last detectors in the congested area upstream of the bottleneck produced the following results: overall, as a consequence of the queue, the flow decreased 5%, which constituted almost the same reductions in all lanes. Lane use was inhomogeneous. The shoulder lane was underused (21%), partly because of the high percentage of trucks in the shoulder lane. The median lane was used intensively by more than 40% of the vehicles. A high percentage of trucks were in the shoulder lane, similarly to the situation at the upstream detector when the VSL system was on. Some trucks chose the center lane, whereas almost no trucks were measured in the median lane. The traffic situation for this category of detectors for the congested period is shown in the left part of Figure 6b.

Second, the first detectors in the free-flowing conditions downstream of the bottleneck were examined. The bottleneck caused an overall flow reduction of 3%, which consisted of a flow gain in the median lane (1%) and flow reductions in the center and the shoulder lanes. The magnitude of the reductions in the center and the shoulder lanes was almost the same. The lane flow distribution remained inhomogeneous, as it was at the detector before, but the shoulder lane was less underused at the cost of the center lane. This result was due to a large increase in the percentage of trucks in the center lane and thus to a reduced flow in that lane. It could be deduced that the trucks changed lanes immediately from the right to the left once they reached free-flowing traffic, although there was no on-ramp in the vicinity of most of the detectors in this category. The traffic situation for this category of detectors is shown in the right part of Figure 6b.

Through a comparison of the results of the two detector locations, driver behavior between the stations, and thus behavior in the

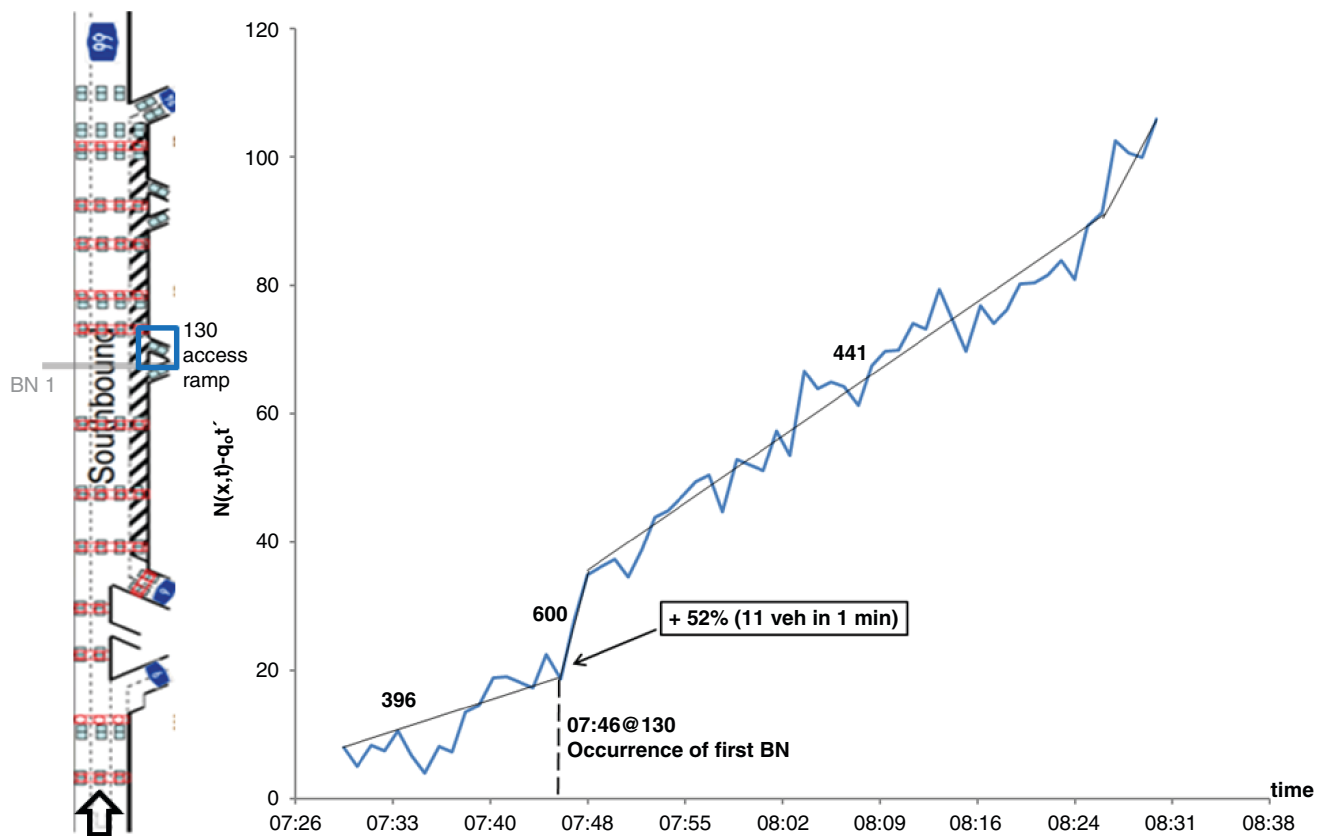


FIGURE 7 On-ramp flow on example day with active VSL system.

transition area between congestion and free flow, could be modeled just as it had been when the VSL system was on. This behavior is illustrated in Figure 6b. When the system was off, drivers were not influenced by VSLs and warnings when they drove out of the queue. They reacted aggressively once they reached free-flowing traffic again. This reaction applied especially to truck drivers. To drive out of the queue thus was as inhomogeneous as it was to drive in congested conditions. Trucks changed immediately to the center lane, and even cars moved to the median lane, which was why there was a gain in capacity in that lane. Fewer vehicles drove in the center lane at the downstream detector because of the higher percentage of trucks there. Drivers chose higher speeds as a consequence of the absence of speed limits. Furthermore, they might have been less afraid of a penalty if they followed at a short distance or, in the case of trucks, if they passed other vehicles. The shorter following distances and the higher speeds might be reasons for the higher flow values compared with the values when the VSL system was operative.

CONCLUSIONS AND OUTLOOK

This study benefited from archived loop detector data available for 25 weekdays with an operative VSL system and for 6 weekdays without an operative VSL system. For the days with the active VSL system, the displayed speed limits and message signs were also known. With the methodology of transformed cumulative curves of vehicle counts and speeds, the spatial-temporal extent of 18 bottlenecks was identified in detail. As a result, traffic flow features such

as congestion form, speed, flow, percentage of trucks per lane, and lane flow distribution were measured on a per lane and total basis.

The macroscopic analysis showed the predominance of wide jams when the system was on, and of stop-and-go traffic when it was off, and seemed to confirm the safety-related aspects of this VSL system. This observation was also confirmed by the lower shock velocities measured in when the system was on, despite uncertainties created by the 1-min data aggregation. A detailed crash and incident analysis was not performed, although one would be valuable in the future.

The analysis of prequeue flow and queue discharge flow showed that the flow drop was homogeneous across all lanes in the presence of a VSL system. In the absence of a VSL system, the flow drop was slightly lower overall, but a small flow gain was observed in the median lane and so were higher flow reductions in the center and shoulder lanes. The VSL system did not solve the problem of under-use of the shoulder lane, but it did successfully harmonize median and center lane flow. The homogenizing and safety effects were observed to cause a slight cost in capacity, however. In addition, the effect of the message sign “Trucks: No Passing” was confirmed by a substantially lower percentage share of trucks in the center lane in the presence of a VSL system. Analysis of the homogeneity between lanes at three detector locations showed notable differences when the system was on and when it was off.

Finally, possible reasons were found for the differences when the system was on and when it was off. The focus was on the behavior of the drivers once they reached free-flowing conditions after they passed through the bottleneck. The drive out of the queue was relatively homogeneous when the system was on. When it was off,

trucks immediately changed to the center lane, and autos even moved to the median lane.

In conclusion, the capacity of this freeway was not increased by the application of the VSL system. In fact, there was even a slight decrease in measured flow. Nevertheless, the safety and homogenizing effects of the VSL system were obvious. Therefore, the total yearly efficiency of the freeway might be higher in terms of avoided accidents and avoided queues. The driver behavior interpretation described earlier could be confirmed through the analysis of further features, including speed distributions, speed variance, and speed compliance. The data sample reported here was not statistically representative. Therefore it would be desirable to have additional data, including data at the individual vehicle level (i.e., floating car data) to conduct a larger study to test the results found here and the assumptions made.

VSL systems are longitudinal systems. They use point-based detector measurements and do not take into account the vehicular behavior between the detectors. For instance, lane changes were not considered sufficiently, and the VSL system might in some cases have introduced unexpected conditions. It was questionable whether those point-measurement inputs were sufficient for the VSL systems and for the analysis. Appropriate methods need to be found to measure lane changes and integrate them into the systems' algorithms by means of an ordinal number.

ACKNOWLEDGMENTS

The authors are grateful to Autobahndirektion Südbayern for the provision of historical data. Steven Boice of DKS Associates and Christian Handke helped to prepare and to process the data.

REFERENCES

- Academic Dictionaries and Encyclopedias, 2010. <http://en.academic.ru/>. Accessed April 2013.
- Straßenverkehrszentrale Baden-Württemberg, 2012. <http://www.svz-bw.de/startseite.html?&L=1>. Accessed May 2013.
- Sisiopiku, V. P. Variable Speed Control: Technologies and Practice. *Proc., 11th Annual Meeting of ITS America*, Miami, Fla., 2001.
- Lee, C., B. Hellinga, and F. Saccomanno. Assessing Safety Benefits of Variable Speed Limits. In *Transportation Research Record: Journal of the Transportation Research Board*, No. 1897, Transportation Research Board of the National Academies, Washington, D.C., 2004, pp. 183–190.
- Bertini, R. L., S. Boice, and K. Bogenberger. Dynamics of Variable Speed Limit System Surrounding Bottleneck on German Autobahn. In *Transportation Research Record: Journal of the Transportation Research Board*, No. 1978, Transportation Research Board of the National Academies, Washington, D.C., 2006, pp. 149–159.
- Bertini, R. L., S. Boice, and K. Bogenberger. Using ITS Data Fusion to Examine Traffic Dynamics on a Freeway with Variable Speed Limits. *Proc., Intelligent Transportation Systems*, 2005, pp. 1006–1011.
- Boice, S., R. L. Bertini, S. Ahn, and K. Bogenberger. Influence of Variable Speed Limit and Driver Information System on Key Traffic Flow Parameters on a German Autobahn. *Proc., 13th ITS World Congress*, London, 2006, pp. 1–8.
- Vukanovic, S. *Intelligent Link Control Framework with Empirical Objective Function: INCA*. PhD dissertation. Technical University of Munich, Germany, 2007.
- van den Hoogen, E., and S. Smulders. Control by Variable Speed Signs: Results of the Dutch Experiment. *Seventh International Conference on Road Traffic Monitoring and Control*, London, 1994, pp. 145–149.
- Papageorgiou, M., E. Kosmatopoulos, and I. Papamichail. Effects of Variable Speed Limits on Motorway Traffic Flow. In *Transportation Research Record: Journal of the Transportation Research Board*, No. 2047, Transportation Research Board of the National Academies, Washington, D.C., 2008, pp. 37–48.
- Hinweise für Steuerungsmodelle von Wechselverkehrszeichenanlagen in Außerortsbereichen. Forschungsgesellschaft für Straßen- und Verkehrswesen e.V. Cologne, Germany, 1992.
- Cassidy, M. J., and R. L. Bertini. Observations at a Freeway Bottleneck. *Transportation and Traffic Theory. Proc., 14th International Symposium on Transportation and Traffic Theory*, Jerusalem, Israel, 1999, pp. 107–124.
- Cassidy, M. J., and R. L. Bertini. Some Traffic Features at Freeway Bottlenecks. *Transportation Research: Part B*, Vol. 33, No. 1, 1999, pp. 25–42.
- Bertini, R. L., and M. J. Cassidy. Some Observed Queue Discharge Features at a Freeway Bottleneck. *Transportation Research: Part A*, Vol. 36, No. 8, pp. 683–697.
- Muñoz, J. C., and C. F. Daganzo. The Bottleneck Mechanism of a Freeway Diverge. *Transportation Research: Part A*, Vol. 36, No. 6, pp. 483–505.
- Cassidy, M. J., and J. R. Windover. Methodology for Assessing Dynamics of Freeway Traffic Flow. In *Transportation Research Record 1484*, TRB, National Research Council, Washington, D.C., 1995, pp. 73–79.
- Bertini, R. L., and A. Myton. Using PeMS Data to Empirically Diagnose Freeway Bottleneck Locations in Orange County, California. Presented at 84th Annual Meeting of the Transportation Research Board, Washington, D.C., 2005.
- Bertini, R. L., S. Hansen, and K. Bogenberger. Empirical Analysis of Traffic Sensor Data Surrounding a Bottleneck on a German Autobahn. In *Transportation Research Record: Journal of the Transportation Research Board*, No. 1934, Transportation Research Board of the National Academies, Washington, D.C., 2005, pp. 97–107.
- Duret, A., S. Ahn, and C. Buisson. Lane Flow Distribution on a Three-Lane Freeway: General Features and the Effects of Traffic Controls. *Transportation Research Part C*, Vol. 24, 2012, pp. 157–167.
- Bogenberger, K., S. Weikl, and R. Bertini. Internal Research Report of the BMW AG on the Categorization of Congestion, 2010.

The Intelligent Transportation Systems Committee peer-reviewed this paper.